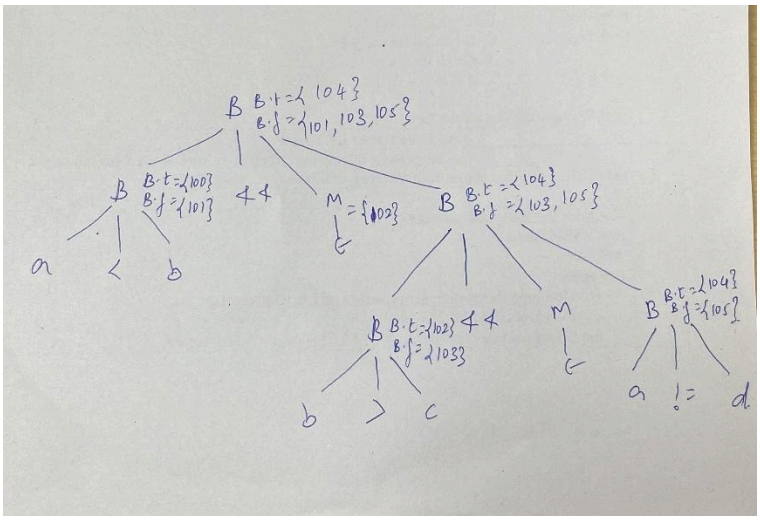
	R V College of Engineering **Department of Computer Science and Engineering** **Improvement CIE: Scheme**		
Subject : (Code)	**COMPILER DESIGN (CS363AI)**	**Semester : 6th BE**	
Date : JUNE 2025	**Duration : 120** minutes	**Staff : DD/SB/SNM/SMS/PK**	
Name :	**USN** :	**Section : A,B,C,D,ISE**	

Si. No	**Part A** **Quiz Questions**	**M**	**BT**	**Co**
1.1	Short circuit codes are the Boolean operators &&, || and ! translations into jumps.---1 mark Example --- 1 mark	2		
1.2	Consider the grammar below: S->SS+ | SS* | a If I ={S->S.S+} compute Closure of (I). Closure(I)={ S->S.S+, S->.SS+, S->.SS*, S->.a}	2		
1.3.	If x<100 goto L2 goto L3 L3: if x>200 goto L4 goto L1 L4: if x!=y goto L2 goto L1 L2: x=2 L1:	2		
1.3	Construct a DAG for the following basic block: T1=A+B T2=C+D T3=E-T2 T4=T1-T3	2		
1.4	Backpatch(p,i)	1		
1.5	Write the backpatching SDT for the following production: B□B1 && M B2 {B.truelist= B2.truelist; B.falselist = merge(B1.falselist,B2.falselist); Backpatch(B1. Truelist, M.instr);}	1		
1.6	It loads into register R1 the value given by contents(contents(100+contents(R2))) and the cost is 3.			
	Part B			

2	Write the different techniques for optimizing basic blocks with examples. 1. The DAG representation of Basic Blocks 2. Finding local common subexpression 3. Dead code elimination 4. The use of Algebraic Identities 5. Representation of array references 6. Pointer assignments and procedure calls Any 5 points with example = 2* 5 = 10 marks	10	L4	CO4

3	Consider the following source code:
	
	for i from1 to 10 do
	for j from 1 to 10 do
	C [i, j]= A[i,j]+B[i,j];
	a) Write the TAC considering the array sizes as 10 X 12 where each element takes 4 bytes of memory space.
	b) Identify the leaders and form the basic blocks
	c) Draw the flow graphs and identify the loops if one exists
	Ans:a) Three address code with properly manipulating the array addresse and jump statements. ---- 4 Marks
	b) Identifying more atleast 6 leaders properly – 3 Marks
	c) Neatly drawing the flow graphs –3 Marks
		4+3
		+3
			L3	CO3

4.	Explain the main issues in code generation with examples for each of the issues in detail. 1. Input to the code generation 2. The Target program 3. Instruction Selection 4. Register allocation 5. Evaluation order Explanation + Examples ----- 2 X5 =10 Marks	10	L2	CO4
5.	How the following basic block can be optimized? 1. a=10 2. b=4*a 3. t1=i*j 4. c=t1+b 5. t2=15*a 6. d=t2*c 7. e=i 8. t3=e*j 9. t4=i*a 10. c=t3+t4 Optimized code: 1. a=10 2. b=4*a 3. t1=i*j 4. c=t1+b 5. t2=15*a	5+5	L3, L2	CO4

	6. d=t2*c 7. t4=i*a 8. c=t3+t4 With proper reason --- 6 Marks Explain with examples Peephole optimization. --- 4 Marks			
6.	Using backpatching manipulate the Truelist and the Falselist at the root node of the annotated parse tree, which should be constructed for the expression: a<b && b>c && a!=d.  	10	L4	CO4